

Agent-Guided Hierarchical Semantic Search over 3D Scene Records

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Abstract

Large visual maps and 3D digital twins contain more scene evidence than a language agent should inspect for every request. We present Agent-MCP Search, an agent-agnostic Model Context Protocol (MCP) for decomposing a natural-language query, traversing a semantic scene hierarchy, validating candidate evidence, and recording a replayable grounding trace. Cursor Agent is the interactive client; a pinned BGE-small encoder supplies reproducible semantic decisions. Lexical search is retained only as a controlled baseline. We also evaluate hierarchy construction rather than treating the hierarchy as a free input: manual, pose-only, pose-plus-semantic, and five-call Cursor-MCP constructions are compared over the same 97 captured views and 150 queries.

On the five-scene query benchmark, lexical graph traversal checks 75.43% fewer views and processes 67.99% fewer input tokens than lexical flat search. Semantic graph traversal checks 79.90% fewer views and processes 32.50% fewer native model tokens than semantic flat search. In both pairs, the hit@1 difference is statistically inconclusive at 95%, while hit@3 declines significantly under the frozen pruning rule. Calibrated fallback on the 48-query ScanNet oracle-map protocol exactly preserves flat lexical R@1/R@3/R@5/MRR while checking 25.0% fewer objects; on the harder 64-query extension it preserves those quality values with 18.8% fewer objects. Validated relation evidence improves ScanNet relational hit@1 from 0.550 to 0.675 with five beneficial and zero harmful top-1 changes. The results support context reduction, not automatic perception or general method superiority.

1 Introduction

Photorealistic 3D representations such as 3D Gaussian Splatting (3DGS) [12] do not by themselves encode that a region is a lobby, an object affords sitting, or a view supports “find the chair farthest from the cabinet.” Object maps, language fields, and scene graphs add this semantic structure, but a second problem remains: a language agent can exceed its useful context budget if it serializes the complete map for each query.

This paper studies semantic search as an agent-map interaction protocol. The map remains local. An agent receives a compact node summary, chooses which branches to expand, validates leaf evidence, and returns stable object/view identifiers. MCP tools separate model reasoning from scene I/O, geometry, and trace persistence. The trace makes every expansion, pruning decision, token count, and fallback auditable.

The hierarchy is not assumed to be automatically correct. We evaluate four ways to construct it and distinguish manual references, agent-assisted maps, official oracle object maps, and native predicted maps. Likewise, Cursor Agent and the reproducible local semantic model are never conflated: Cursor provides an interactive product demonstration and direct hierarchy decisions, whereas a pinned BGE encoder executes all 150 query-time experiments. Deterministic lexical graph and flat search remain controls that isolate candidate traversal.

Contributions.

- An agent-agnostic MCP protocol for query decomposition, top-down hierarchy traversal, grounded evidence validation, and complete replayable traces over 3D scene records.
- A controlled four-variant comparison of flat/graph lexical and flat/graph neural semantic retrieval on identical scenes, records, queries, and labels, with native tokenizer counts, latency, calls, and paired confidence intervals.
- A four-way hierarchy-construction evaluation covering manual, pose-only, pose-plus-semantic, and Cursor-MCP maps, including structural agreement, query quality, context, construction time, and missing usage provenance.
- Stronger public protocols with explicit total, positive, negative, and box-eligible denominators, harder functional/relational queries, calibrated fallback, and a controlled relation audit.
- A release contract in which every table and figure is generated or checked against saved per-query ev-

idence, with oracle and predicted-map results kept protocol-separated.

2 Related Work

Hierarchical language agents and scene graphs. SayPlan [18] is the closest prior to our query protocol: an LLM expands and contracts a collapsed hierarchical 3D scene graph to obtain a task-relevant subgraph before planning. Our scope is narrower but more instrumented: semantic retrieval and grounding rather than long-horizon robot planning, an MCP tool boundary, same-map graph-vs.-flat controls, and complete per-query cost traces. HOV-SG and Hydra organize semantic/spatial information for navigation and robotics [21, 9]. Search3D constructs a hierarchical open-vocabulary representation spanning parts, objects, and regions [20]; we instead evaluate hierarchy as a query-context controller over frozen scene records.

Open-vocabulary 3D maps and grounding. ConceptFusion and OpenScene build open-set or open-vocabulary 3D representations [10, 16]. ConceptGraphs constructs object-centric maps from RGB-D observations [7]. Beyond Bare Queries (BBQ) combines object captions, 3D extents, relations, and deductive reasoning for open-vocabulary grounding [15]; it motivates our ScanNet/Replica scene and metric alignment. We do not report an official BBQ run and make no superiority claim.

Language fields and semantic Gaussians. LERF [13] embeds language in a radiance field. LangSplat, Semantic Gaussians, and LEGS attach open-vocabulary features to Gaussian representations [17, 8, 22]. These methods address representation learning and dense relevance. Our primary variable is the amount of an existing semantic map exposed to a query client.

Datasets and evaluation. ScanNet provides annotated RGB-D reconstructions [4], while Replica provides high-quality indoor replicas [19]. Nr3D and Sr3D support referring expression grounding in ScanNet scenes [1]; ScanRefer evaluates 3D referring expressions [2]. OpenLex3D advocates tiered evaluation of open-vocabulary 3D representations [11]. We therefore separate view retrieval, oracle-map object retrieval, 3D box accuracy, and native predicted-map execution.

Context efficiency. Prompt compression and retrieval pruning remove irrelevant textual evidence [14, 5, 3, 6]. Tokens alone do not establish grounding quality; we report context, checked candidates, retrieval quality, and geometry together.

3 Method

3.1 Agent-MCP Query Protocol

Let a scene hierarchy be $H = (V, E)$ and let $\mathcal{C}$ be canonical semantic records linked to its leaves. A query client does not receive $\mathcal{C}$ in one prompt. It uses the following protocol:

1. decompose query q into target, attributes, optional anchor/relation, and intent;
2. request the root and compact child summaries through MCP;
3. select conservative child IDs to expand and record rejected siblings;
4. retrieve candidate leaf views/objects and validate semantic or spatial evidence;
5. rank stable IDs, return supporting evidence, and finalize a trace.

The MCP server owns scene access, hierarchy inspection, geometry, and trace persistence. The client owns semantic decisions. This boundary permits Cursor Agent, a local model, or another compatible client without changing map storage. The trace stores exact tool requests/responses, model identity, branches considered, token method, latency, fallback, and final IDs.

Clients and controls. Cursor Agent is the interactive client used for direct MCP hierarchy construction. Its UI label in this study is “Cursor Grok 4.5 High Fast.” Cursor did not expose auditable token usage, so those fields remain unavailable. Reproducible query experiments use BAAI/bge-small-en-v1.5, a 384-dimensional encoder run with FastEmbed ONNX on CPU. It replaces free-form agent choices with cosine similarity while preserving the same top-down decision boundary. It is a semantic model, not an LLM and not evidence of Cursor quality. Lexical overlap is a non-model control.

3.2 Hierarchy Construction

Each captured record contains a posed view, semantic summary, objects, regions, landmarks, attributes, relations, and optional boxes. Construction produces scene-zone-entity-view containment edges and stable node IDs. We evaluate four variants:

1. Manual reference: existing human/Cursor-assisted zones;
2. Pose only: deterministic camera-position clustering with spatial scale factor 3.5;
3. Pose + semantic: pose clusters followed by deterministic summary-based merging;

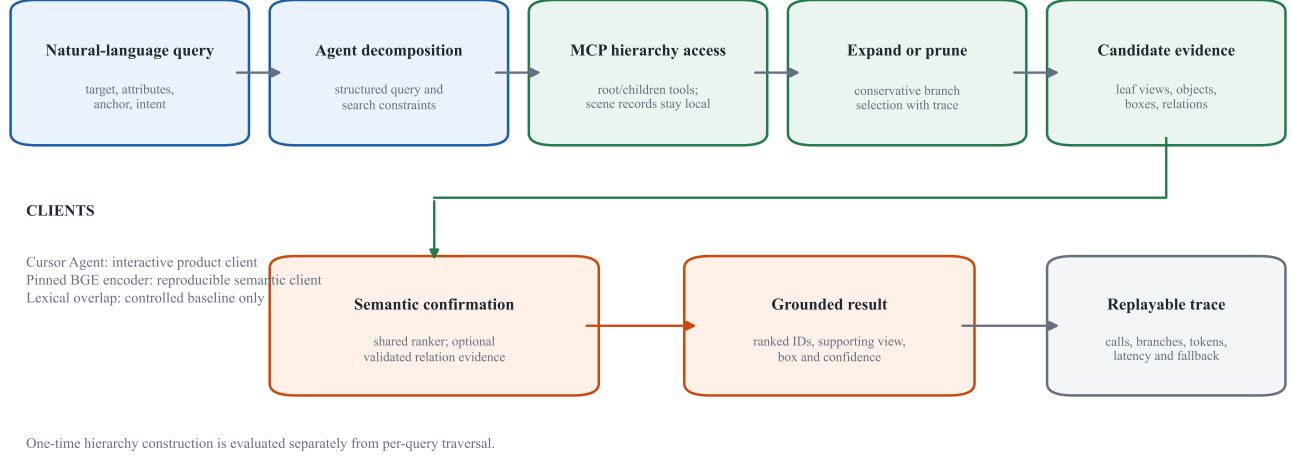


Figure 1: Agent–MCP Search. The agent decomposes a query, reads local hierarchy nodes through MCP, expands or prunes branches, validates leaf evidence, and writes a replayable trace. Cursor Agent is the interactive client; the pinned local encoder supplies the reproducible semantic client. One-time hierarchy construction is evaluated separately.

4. Cursor Agent + MCP: five direct `build_spatial_clusters` calls, followed by agent zone decisions.

The Cursor client was not shown manual-zone labels, did not call persistence tools, and did not modify scene data. Its five input cluster counts were 1, 1, 1, 2, 3 and its final hierarchy covers all 97 views without duplicate assignment. Fixed thresholds, seed 20260724, decisions, and rationales are released. Manual zones overlap by design, so structure agreement uses multi-membership pairwise F1 and weighted best-zone Jaccard rather than a single-label clustering metric.

3.3 Semantic Records and Structured Queries

Each searchable candidate is

$$c_i = (\text{id}_i, \tau_i, A_i, y_i, x_i, R_i, B_i, V_i), \quad (1)$$

where τ_i is type, A_i the ancestor path, y_i label, x_i searchable text, R_i typed relations, B_i optional 2D/3D boxes, and V_i supporting views. Source object IDs are retained where available. Graph and flat variants consume identical c_i .

A query is represented as

$$q = (q_t, q_a, r, \mathcal{A}, \iota), \quad (2)$$

with target q_t , anchor q_a , relation r , attributes $\mathcal{A}$, and intent ι . Intent expansion maps functional requests to candidate affordances; for example, “find a place to sit” activates chair, sofa, bench, couch, armchair, lounge, and seating records.

3.4 Top-Down Semantic Traversal

At node v , the semantic client embeds q and each child summary. With cosine scores z_j , it retains up to three children whose score is at least 0.20 and within 0.08 of the best child. If no child meets the threshold, a fixed three-child fallback is recorded. The same encoder ranks leaf evidence. Flat-semantic search embeds all scene records without hierarchy decisions.

The lexical graph control computes branch token overlap u_j . If $u_{\max} > 0$, branches satisfying

$$u_j \geq \max(1, 0.5u_{\max}) \quad (3)$$

are retained; if all overlap is zero, all branches are eligible. The identical lexical scorer ranks retained or flat candidates. Stable-ID tie breaking keeps the run deterministic.

3.5 Validated Relation Reasoning

Relations are parsed only when a target–anchor predicate and required frame exist. Anchor confidence and margin are checked before geometry. Valid 3D boxes are preferred; normalized 2D boxes are used only for image-frame relations. Non-finite coordinates, non-positive extents, or absent viewpoint frames invalidate evidence.

Relation evidence may change ranking only if confidence is at least 0.65 and the candidate’s lexical score is within 0.15 of the best lexical candidate. Unresolved evidence triggers deterministic fallback and cannot break a lexical tie. This prevents world-coordinate boxes from

masquerading as viewpoint-relative left/right/front/behind evidence.

3.6 Output, Accounting, and Complexity

Every result contains ranked IDs, supporting views/boxes, visited ancestors, checked counts, fallback reason, runtime, serialized context, token method, model calls, GT eligibility, and provenance. Model input tokens use the native BGE tokenizer; lexical context uses a pinned local tokenizer. Cursor usage is reported only when exposed. One-time construction cost is never mixed with per-query cost.

For N scene records and N_q retained records, flat scoring is $O(N)$ and graph leaf scoring is $O(N_q)$ after the hierarchy gate. Fallback has the flat worst case. An agent may make more small calls while processing fewer total tokens; both values are therefore reported.

4 Datasets and Evaluation

4.1 Evaluation Tracks

Captured scenes. Five team-captured indoor/outdoor scenes contain 97 RGB-D views and 1,066 semantic items. The benchmark has 25 queries in each of six categories: simple-object, attribute, relational, functional, multi-hop, and negative. Verified expected views exist for 125 queries. The remaining 25 negative queries contribute to cost only because no independent negative GT exists.

Public oracle-map tracks. The BBQ-aligned Replica subset contains 575 official object boxes. The ScanNet subset contains 392 official boxes and 48 mapped Nr3D/Sr3D+ queries. The extended ScanNet track adds one GT-backed functional query and one verified-absence query per scene. These tracks evaluate retrieval over known objects, not automatic segmentation or map construction.

Native predicted-map tracks. ConceptGraphs constructs predicted maps for all eight selected ScanNet and eight Replica scenes. LangSplat executes RGB 3DGS optimization, SAM/CLIP preprocessing, autoencoder compression, language-field optimization, rendering, and six queries on ScanNet 0011_00. Different map construction and resource profiles keep these executions outside the oracle-map ranking.

4.2 Metrics and Statistical Protocol

For each method/query we record views V , objects O , nodes G , elapsed time t , context characters C , input tokens T , and model calls. Ratio-of-sums reduction against

flat is

$$\Delta_X = 100 \left(1 - \frac{\sum_q X_{gq}}{\sum_q X_{fq}} \right) \%. \quad (4)$$

This definition yields 75.43% fewer views for lexical graph traversal and 75.4% when rounded to one decimal.

For ranked expected views or objects we report hit/Recall@ k and reciprocal rank. A 3D prediction is scored with

$$\text{IoU}_{3D} = \frac{|B_p \cap B_g|}{|B_p \cup B_g|}, \quad (5)$$

$$\text{Acc@}\tau = \frac{1}{|Q_{\text{box}}|} \sum_q \mathbf{1}[\text{IoU}_{3D} \geq \tau]. \quad (6)$$

Eligible misses score zero. Segmentation mIoU is unavailable without paired predicted and reference point labels. Paired differences use 10,000 query bootstrap resamples with seed 20260715.

5 Experiments

5.1 Four Query Variants

Lexical graph reduces views by 75.43% and input tokens by 67.99%. Its graph-minus-flat hit@1 difference is -0.088 with 95% CI $[-0.184, 0.008]$, which is inconclusive. The hit@3 difference is -0.120 with CI $[-0.184, -0.056]$, a significant decline.

Semantic graph reduces views by 79.90% and native input tokens by 32.50%. Mean per-query token reduction is 32.73% with CI $[27.11, 38.21]$. The hit@1 difference is -0.016 with CI $[-0.096, 0.064]$ and is inconclusive. The hit@3 difference is -0.144 with CI $[-0.224, -0.072]$; MRR differs by -0.082 with CI $[-0.150, -0.016]$. Thus the frozen semantic gate saves context and CPU time but does not preserve top-3 quality.

Graph-semantic execution makes more small encoder calls because it scores successive hierarchy levels. This is why call count and processed tokens are reported separately. Model load is 9.459s; observed process RSS after load is 212.88 MB and peak observed RSS is 471.00 MB.

5.2 Hierarchy Construction

Pose-plus-semantic gives the highest hit@ k in this experiment. Cursor-MCP is structurally closer to the manual reference than either deterministic alternative and uses less query context, but does not dominate their quality. The correct claim is therefore agent-assisted hierarchy construction, not fully automatic construction.

All deterministic variants consume zero provider tokens. Historical manual time and Cursor UI runtime/-token usage are unavailable, not zero. Each variant has 8,939 locally estimated serialized input-context tokens. The Cursor result is supported by five direct MCP calls and saved rationales; it is not inferred from the manual hierarchy.

Table 1: Evaluation tracks and denominators. Q_{total} contributes to cost; Q_+ to retrieval quality; Q_- to negative accuracy; Q_{box} to 3D box metrics.

Track	Scenes	Q_{total}	Q_+	Q_-	Q_{box}	Map/evidence type
Captured scenes	5	150	125	0	0	Manual records and verified expected views
Replica	8	56	48	8	48	Official oracle object IDs and boxes
ScanNet original	8	48	48	0	48	Official oracle object IDs/boxes; Nr3D/Sr3D+
ScanNet extended	8	64	56	8	56	Original plus functional and verified-absence queries
ConceptGraphs native	16	104	96	0	96	Predicted object maps; separate protocol
LangSplat native	1	6	6	0	0	Learned language field; separate protocol

Table 2: Metric prerequisites and denominators. Missing evidence remains unavailable rather than becoming zero.

Metric	Prediction requirement	Reference requirement	Denominator
Efficiency and cost			
Checked views/objects/nodes	Traversal trace	None	All completed queries
Runtime and usage	Timed result and count metadata	None	Results with that measurement
Retrieval quality			
hit@ k	Ranked view IDs	Verified expected view IDs	Eligible positive-view queries
Recall@ k , MRR	Ranked object IDs	Official expected object IDs	Object-ID-eligible positive queries
Negative accuracy	Explicit not-found result	Verified object absence	Verified negative queries
Geometry and segmentation			
3D IoU, Acc@ τ	Valid predicted 3D box	Valid official 3D box	Box-eligible queries; misses score zero
Segmentation mIoU	Predicted point labels	Paired GT segmentation	Paired points/classes only

5.3 Public Oracle-Map Protocols

Replica quality saturates because the oracle labels are lexically easy. It is a sanity check, not evidence that variants are equivalent generally. Graph+fallback preserves the flat lexical ceiling while checking 27.9% fewer objects.

On original ScanNet, graph+fallback exactly preserves flat lexical R@1, R@3, R@5, MRR, and Acc@0.25 while checking 25.0% fewer objects and using 24.9% fewer estimated tokens. Aggressive graph search checks 90.3% fewer objects but loses 2.1 points at R@1, R@3, and R@5. On extended ScanNet, fallback again preserves flat values with 18.8% fewer objects and 18.7% fewer tokens.

The extended positive denominator is 56, not 64; the remaining eight queries measure verified negative accuracy. Its eight functional queries obtain R@1=.625, R@3=.750, R@5=.750, and MRR=.667 under graph+fallback. Attribute disambiguation remains weak (two queries; R@1=.000).

5.4 Controlled Relation Audit

The relation audit compares graph+fallback with the identical pipeline under no_relation=true. On 40 ScanNet relation-labelled queries, 35 parse to a supported

target-anchor predicate and five remain explicitly unparsed. Validated relations increase hit@1 from .550 to .675. Exactly five top-1 changes are beneficial, zero are harmful, and all five pass the confidence and score-margin gates. Gains comprise one between, one closest, and three farthest queries.

The prior relation underperformance arose because low-confidence evidence could break lexical ties and frame-dependent predicates could be applied without adequate validation. The new policy blocks both behaviors. On Replica, eight parsed near queries and eight sequential multi-hop prompts remain correct with and without reranking; no top-1 result changes.

5.5 Native External Executions

ConceptGraphs produces predicted object maps on all 16 selected public scenes. Its Acc@0.25 is .0625 on both dataset tracks. LangSplat records a complete native pipeline execution and selects an expected view for four of six queries, but returns relevance points rather than 3D boxes. Its hardware-adapted SAM-B/CLIP-B run uses 3,000 RGB and language iterations on a 6 GB GPU, not the original 30,000-iteration/24 GB scale.

External token totals use each native local tokenizer and are not comparable to character-derived context es-

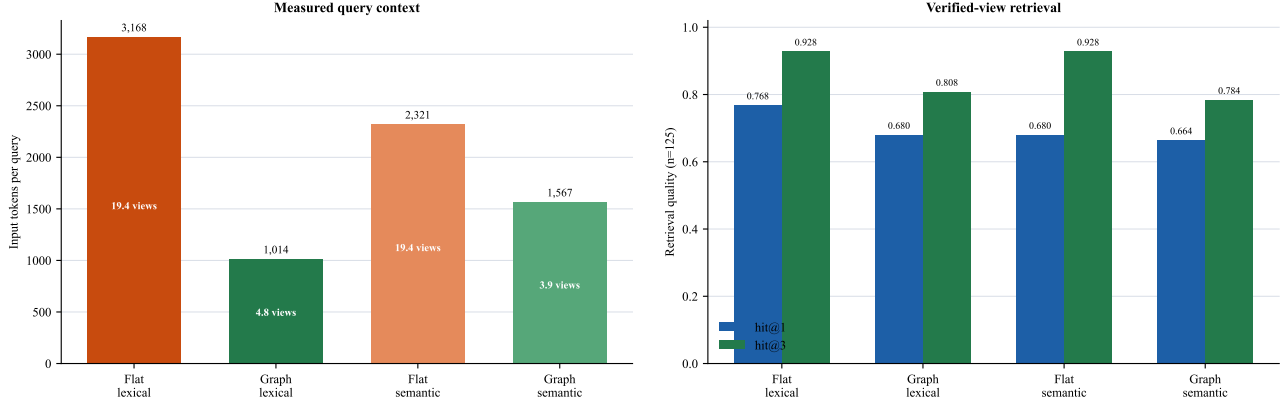


Figure 2: Four query variants over identical scenes, records, queries, and quality labels. Semantic tokens use the model’s native tokenizer; lexical tokens count serialized inspected context.

Table 3: Same-input query benchmark. Quality uses $Q_+ = 125$; cost uses $Q_{\text{total}} = 150$. Runtime includes local model execution where applicable.

Method	hit@1	hit@3	MRR	Obj. hit	Views	Tokens	ms	Calls	Total tokens
Flat lexical	.768	.928	.846	.648	19.40	3,168	6.39	0.00	475,230
Graph lexical	.680	.808	.742	.632	4.77	1,014	5.53	0.00	152,139
Flat semantic	.680	.928	.800	.640	19.40	2,321	872.42	3.00	348,209
Graph semantic	.664	.784	.717	.624	3.90	1,567	364.19	8.25	235,045

timates or API billing. No official BBQ execution artifact is included.

5.6 Qualitative Trace

For “Find the piano,” the released local-semantic trace records query encoding, root scores, three retained zone branches, leaf-level scores, supporting views, native tokens, and per-step latency. It is reloadable under the same trace schema used by Cursor. The local trace demonstrates reproducibility, not a Cursor query result.

6 Discussion

The experiments answer three separate questions. First, hierarchical traversal reduces context for both lexical and neural semantic scorers. Second, the current hard pruning policy does not preserve top-3 retrieval quality on the captured-scene benchmark; calibrated fallback is therefore necessary. Third, hierarchy construction materially affects both cost and quality. A pose-plus-semantic hierarchy can outperform the manual reference on $\text{hit}@k$ while differing structurally, and the Cursor-MCP hierarchy trades some quality for closer reference agreement and lower context.

Public results reinforce the need for protocol-specific wording. Replica is a ceiling sanity check. ScanNet dis-

tinguishes aggressive pruning, calibrated fallback, and flat search. The safe conclusion is that fallback preserves flat lexical quality on these two frozen ScanNet protocols, not that graph routing universally preserves flat retrieval. Native ConceptGraphs and LangSplat runs verify integration under predicted/learned maps but cannot establish a ranking against oracle-map retrieval.

The controlled relation result also changes the earlier diagnosis. Relation reasoning itself is not harmful after anchor, frame, geometry, confidence, and margin checks. The remaining failures are unsupported language forms and attribute ambiguity, which should be addressed by improved parsing or a stronger instruction-tuned client rather than by weakening geometric safety.

7 Limitations

- Cursor Agent supplies direct hierarchy-construction decisions, but the 150-query benchmark uses a pinned embedding model rather than Cursor or an instruction-tuned LLM. No Cursor query-quality claim is made.
- Cursor UI provider tokens and wall-clock construction time were not exposed. They are unavailable, not zero; only serialized local input context is measured.

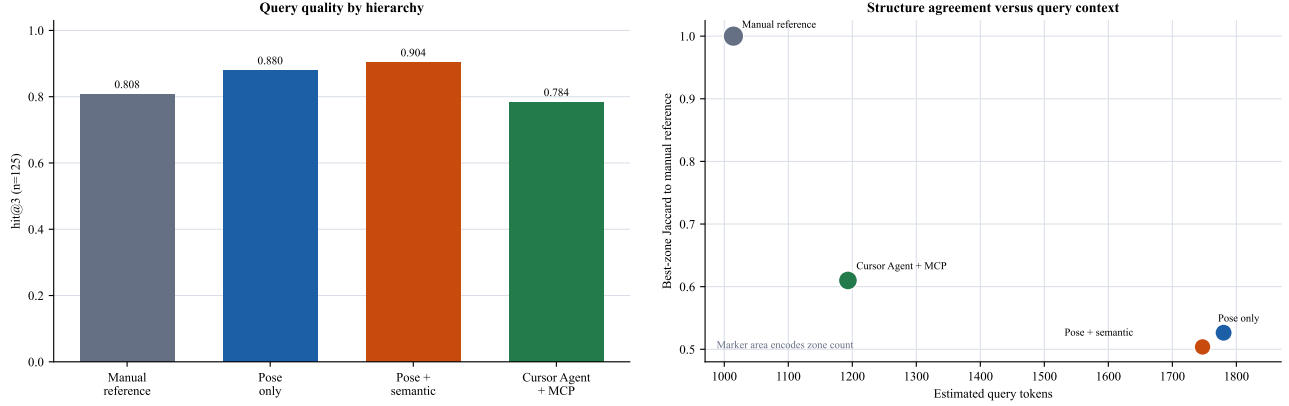


Figure 3: Hierarchy construction changes grouping and therefore query behavior, but not underlying semantic records. Manual agreement is a project reference, not independent GT.

Table 4: Hierarchy construction on the same 97 views and 150 queries. Agreement compares to overlapping manual zones; quality uses $n = 125$.

Hierarchy	Zones	Pair F1	Jacc.	hit@1	hit@3	MRR	Views	Tokens	Build s
Manual reference	26	1.000	1.000	.680	.808	.742	4.77	1,014	N/A
Pose only	10	.588	.526	.760	.880	.829	6.07	1,780	.042
Pose + semantic	8	.570	.504	.784	.904	.845	5.80	1,747	.042
Cursor Agent + MCP	18	.616	.610	.704	.784	.749	5.53	1,193	N/A

- Captured-scene ViewJSON and expected views are manual reference artifacts. Their quality can contain correlated annotation bias.
- The best construction variant is still agent-assisted and starts from manual semantic records. This paper does not establish automatic semantic perception.
- Replica and ScanNet retrieval use oracle object identities and boxes. Their Acc@0.25 is retrieval-over-GT-map accuracy, not predicted-map segmentation or localization accuracy.
- Semantic graph pruning significantly reduces hit@3 and MRR under the frozen policy. Quality-constrained calibration remains the central algorithm target.
- Native external runs differ in maps, frame sampling, output geometry, and resources. They are execution evidence, not fair comparative rankings.
- LangSplat covers one scene under a reduced-resource profile, and no official BBQ execution is included.

8 Reproducibility and Release

The release includes method and trace schemas, 600 per-query rows for the four query variants, 300 semantic traces, four hierarchy constructions, per-query hierarchy results, three public protocols, relation audits, run configurations, model/checkpoint provenance, seeds, and hardware metadata. All 150 queries finish under all four internal methods; validation reports contain zero errors.

The semantic run records Windows 11, Python 3.13.13, FastEmbed 0.8.0, ONNX Runtime 1.27.0, and the BGE model name. ScanNet and Replica raw files are not redistributed; official access and import manifests are provided. Figure scripts read the same frozen JSON checked by the paper audit.

```
python -B -m pytest -q -p no:cacheprovider tests
python -B scripts/validate_agent_semantic_benchmark.py \
  --output outputs/agent_semantic/five_scene_four_variant_v1
python -B scripts/check_academic_paper.py
papers/beyond-proximity/build_paper.cmd
```

9 Conclusion

Agent-MCP Search restores the original agent-centered idea as an explicit, auditable MCP protocol rather than a hidden lexical heuristic. Cursor Agent is a demonstrated interactive client and hierarchy constructor; a pinned local semantic model provides reproducible query evidence.

Table 5: Public oracle-map results. Replica is a lexical sanity ceiling. ScanNet original and extended are discriminative. Tokens are local context estimates, not provider billing.

Track	Method	R@1	R@3	R@5	MRR	A@.25	Neg.	Objects	Tokens
Replica, $Q = 56$	Graph	1.000	1.000	1.000	1.000	1.000	1.000	12.11	303
	Graph+fallback	1.000	1.000	1.000	1.000	1.000	1.000	51.82	1,275
	Flat lexical	1.000	1.000	1.000	1.000	1.000	1.000	71.88	1,765
ScanNet, $Q = 48$	Graph	.604	.813	.854	.718	.604	N/A	4.73	219
	Graph+fallback	.625	.833	.875	.743	.604	N/A	36.75	1,589
	Flat lexical	.625	.833	.875	.743	.604	N/A	49.00	2,117
ScanNet ext., $Q = 64$	Graph	.607	.786	.821	.705	.589	1.000	4.38	200
	Graph+fallback	.625	.821	.857	.732	.589	1.000	39.81	1,720
	Flat lexical	.625	.821	.857	.732	.589	1.000	49.00	2,115

Table 6: Controlled ScanNet relation audit ($Q = 40$).

Measure	No relation	Validated relation
hit@1	.550	.675
Parsed / unparsed	35 / 5	
Beneficial changes	0	5
Harmful changes	0	0
Validated changed winners	0	5

Table 7: Native external execution evidence. Protocols differ from the oracle-map experiments; this is not a leaderboard.

Execution	S	Q	Quality	s/q	Tokens
CG-ScanNet	8	48	A@.25=.063	.336	627
CG-Replica	8	56	A@.25=.063	.274	459
LangSplat-ScanNet	1	6	view hit=.667	2.037	74

Across identical five-scene inputs, hierarchy reduces lexical and semantic query context substantially, but the significant hit@3/MRR losses show that uncalibrated hard pruning is not yet quality preserving. Calibrated fallback preserves flat lexical R@1/R@3/R@5/MRR on the two frozen ScanNet protocols while reducing objects and context, and validated relations improve relational hit@1 without harmful top-1 changes. The next scientific step is an instruction-tuned, quality-constrained agent evaluation on predicted public maps under one shared scene/-query/resource protocol.

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